



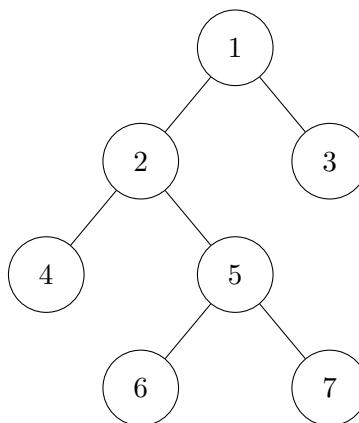
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******* INSTRUCTIONS *******

- For **Question 1**, construct the binary tree as shown in the diagram and ensure that all required interfaces and abstract classes are properly included.
- Ensure that the printed output clearly reflects the structure of the tree and matches the specified format.
- Add clear screenshots of your program output corresponding to **Questions 1 and 2**.
- All submitted work must be your own (or your group's) original work. Academic integrity policies apply.

Question 1: Create a Binary Tree

Complete the main method (given below) in which you have to complete a **Binary Tree** using the `LinkedBinaryTree` class that we studied during the lecture. All the required interfaces and classes have been uploaded on the Brightspace. Make sure that you include all the required interfaces and abstract classes. The structure of the tree should look like this:



```

1 public static void main(String[] args) {
2     // Create a new binary tree
3     LinkedBinaryTree<Integer> binaryTree = new LinkedBinaryTree<>();
4     // Adding the root element (1)
5     Position<Integer> root = binaryTree.addRoot(1);
6
7     // Adding left and right children (2 and 3) to the root
8     Position<Integer> leftChild = binaryTree.addLeft(root, 2);
9     Position<Integer> rightChild = binaryTree.addRight(root, 3);
10
11     //Your code should come here
12 }

```

After creating a complete Binary Tree, your main method should be able to print the elements of the tree. Therefore, the output should look like this:

Output:

```

Binary Tree Structure:
Root: 1
Left Child of Root: 2
Right Child of Root: 3
Left Child of 2: 4
Right Child of 2: 5
Left Child of 5: 6
Right Child of 5: 7

```

Question 2: Inorder Traversal of a Binary Tree

Modify your main method so that the elements are now printed using the inorder tree traversal of the above Binary Tree. It is worth mentioning here that the methods required for the inorder tree traversal have already been included in the provided `AbstractBinaryTree` class.

Hint: You have to call the `inorder()` method to get an iterable collection of positions of the tree (reported in inorder) and then printout using the for-each loop with `getElement()` method of the provided `Position` interface.

The output should look like this:

Output:

```

Inorder Tree Traversal:
4 2 6 5 7 1 3

```